INCH-POUND
MIL-PRF-62411/1A(AT)
7 July 1997
SUPERSEDING
MIL-H-62411/1(AT)
5 July 1985

PERFORMANCE SPECIFICATION SHEET

HEAT EXCHANGER, NBC SYSTEM
HEAT EXCHANGER P/N M62411/1
(ENGINEERING P/N 12336771)

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-PRF-62411A(AT) listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS

Design and construction. The heat exchanger shall conform to drawing hierarchy of which Drawing 12336771 is the top drawing.

Weight. The weight of the heat exchanger shall not exceed 14 pounds (lb) (6.35 kilograms (kg)) (see 4.6.3).

Performance.

Leakage.

External leakage. The pneumatic pressure applied to the hot air passages of the heat exchanger shall be 62 pounds per square inch gage (psig) (427 kilopascals (kPa)) (see 4.6.4.1).

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Internal leakage. The pressure applied to the hot air passages shall be 62 psig. The pressure loss in the first 20 seconds shall not exceed 25 psig (172 kPa) (see 4.6.4.2).

Proof pressure. The heat exchanger shall withstand a proof pressure of 70 psig (483 kPa)) (see 4.6.5).

Burst pressure. The heat exchanger shall withstand a burst pressure of 117 psig (807 kPa) at 360 degrees Fahrenheit (°F) (182 degrees Celsius (°C)) (see 4.6.6).

Heat transfer. Heat transfer performance shall be not less than that shown in figure 1 (see 4.6.7).

Air pressure loss. The pressure loss shall not exceed that shown in figure 2 (see 4.6.8).

Leakage.

External leakage. A pneumatic pressure of 62 to 64 psig (427 to 441 kPa) shall be applied to the hot air passages.

Internal leakage. A pneumatic pressure of 62 to 64 psig shall be applied to the hot air passages.

Proof pressure. The test pressure shall be 70 to 73 psig (483 to 503 kPa).

Burst pressure. The oven temperature shall be 360°F (182°C). The test pressure shall be 117 to 121 psig (807 to 834 kPa).

Heat transfer. To determine conformance to 3.5.4, the heat exchanger shall operate at each combination of the steady state conditions specified in table VI. The measured hot air outlet temperature shall be equal to or greater than that specified in table VI (see figures 1 and 2).

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TABLE VI. Heat transfer.

Initial air temperature °F (°C)		Air flow lb/min (kg/min)		Hot air outlet outlet temperature
Hot	Cold	Hot	Cold	°F ±10°F (°C ±5.6°C)
210 (99)		10 (4.5)	15 (6.8)	174 (79)
210			30 (13.6)	200 (93)
210			52.2 (23.7)	206 (97)
210		14.7 (6.7)	15	150 (66)
210			30	190 (88)
210			52.2	202 (94)
210		20 (90.7)	15	147 (64)
210			30	180 (82)
210			52.2	202 (94)

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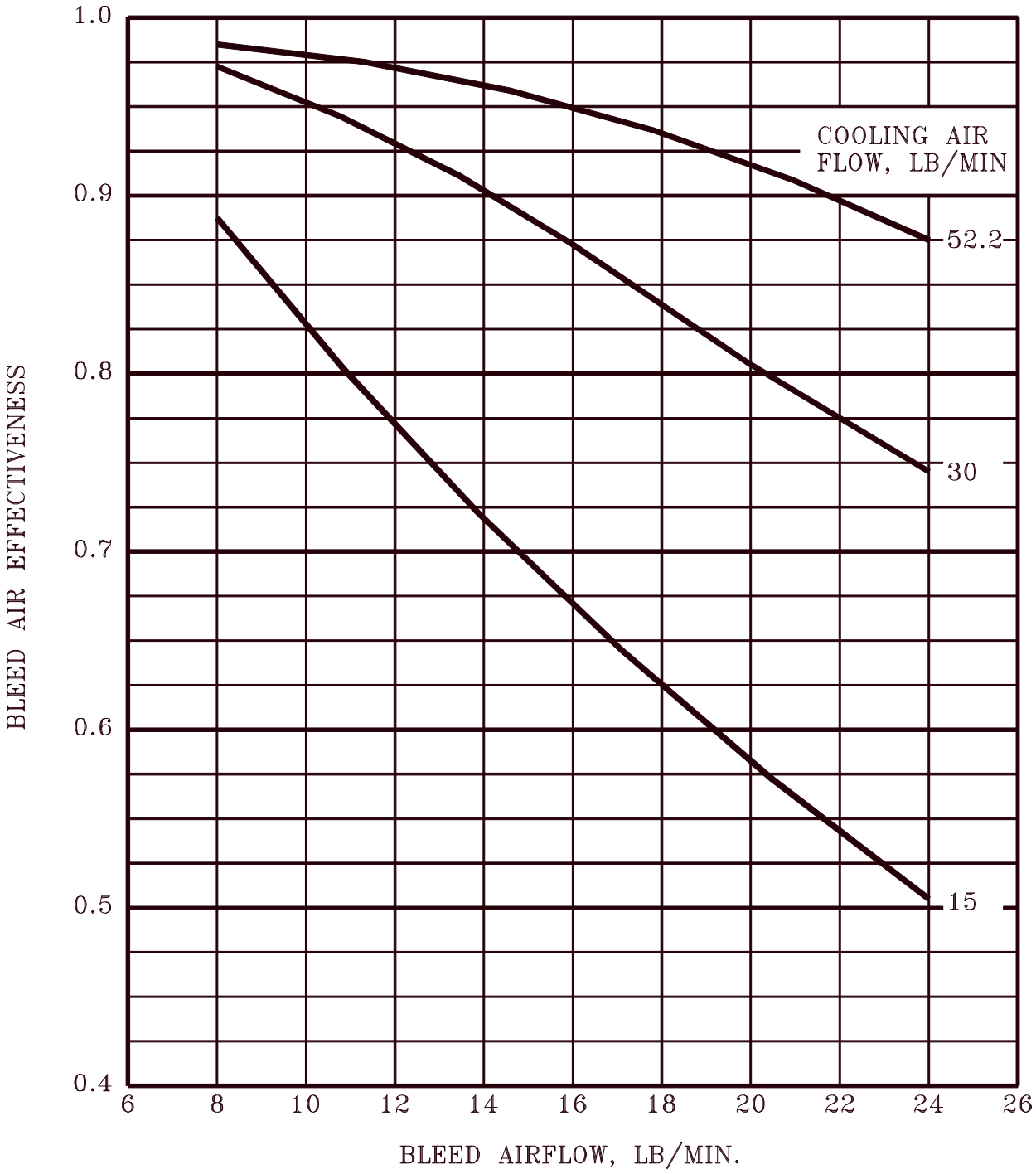


FIGURE 1. Required heat exchanger performance.

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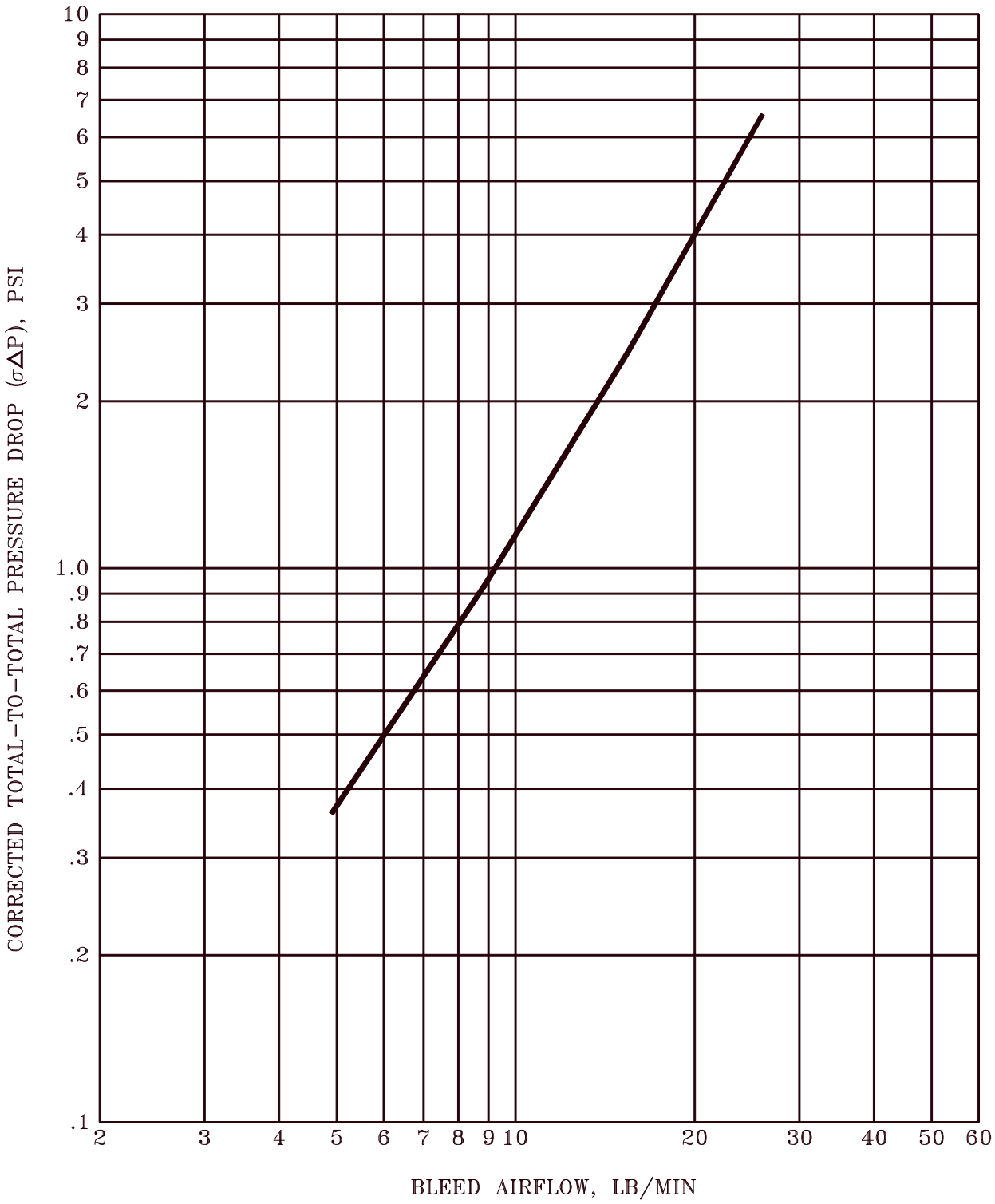


FIGURE 2. Allowable heat exchanger bleed air pressure loss.

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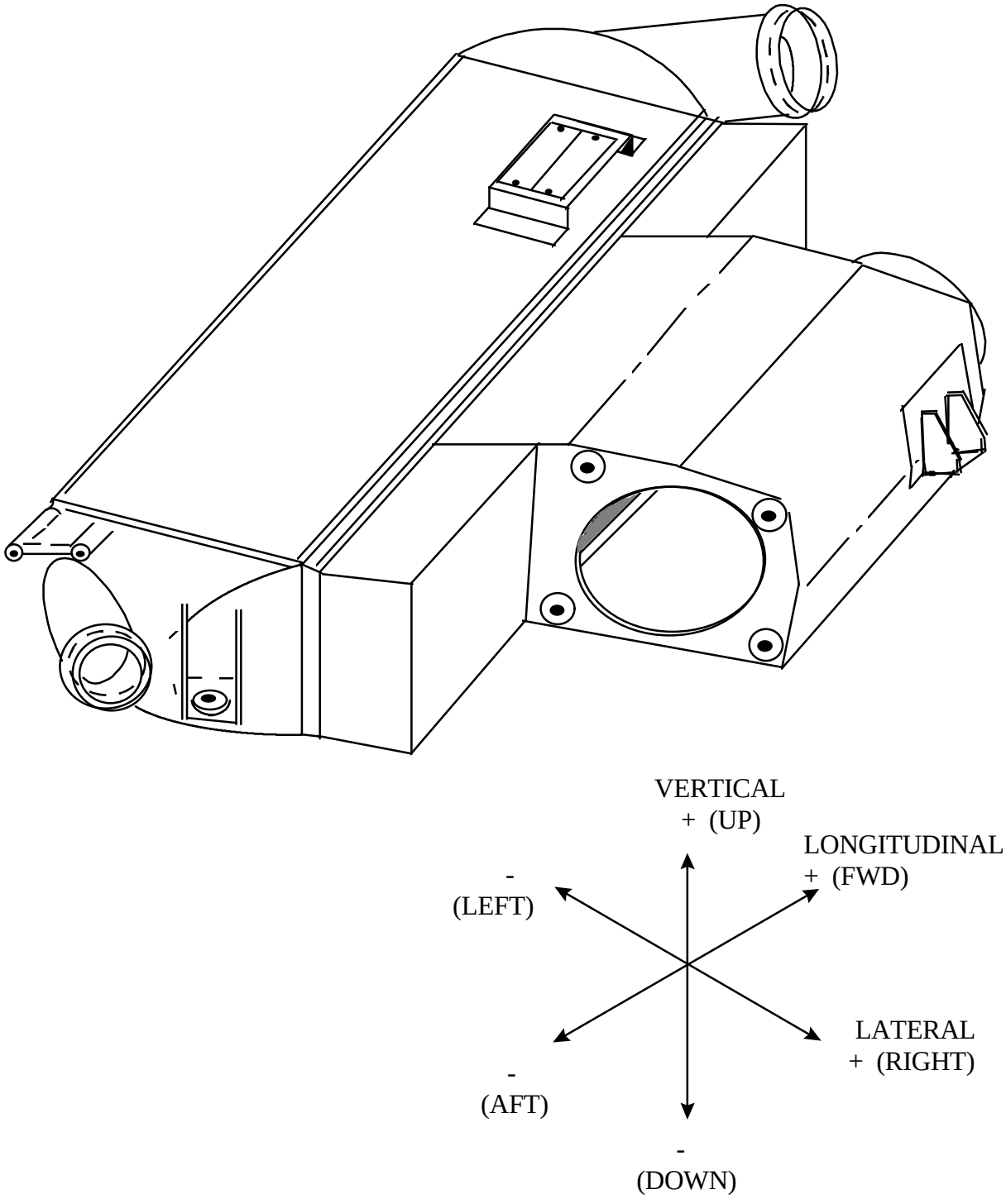


FIGURE 3. Axes orientation.

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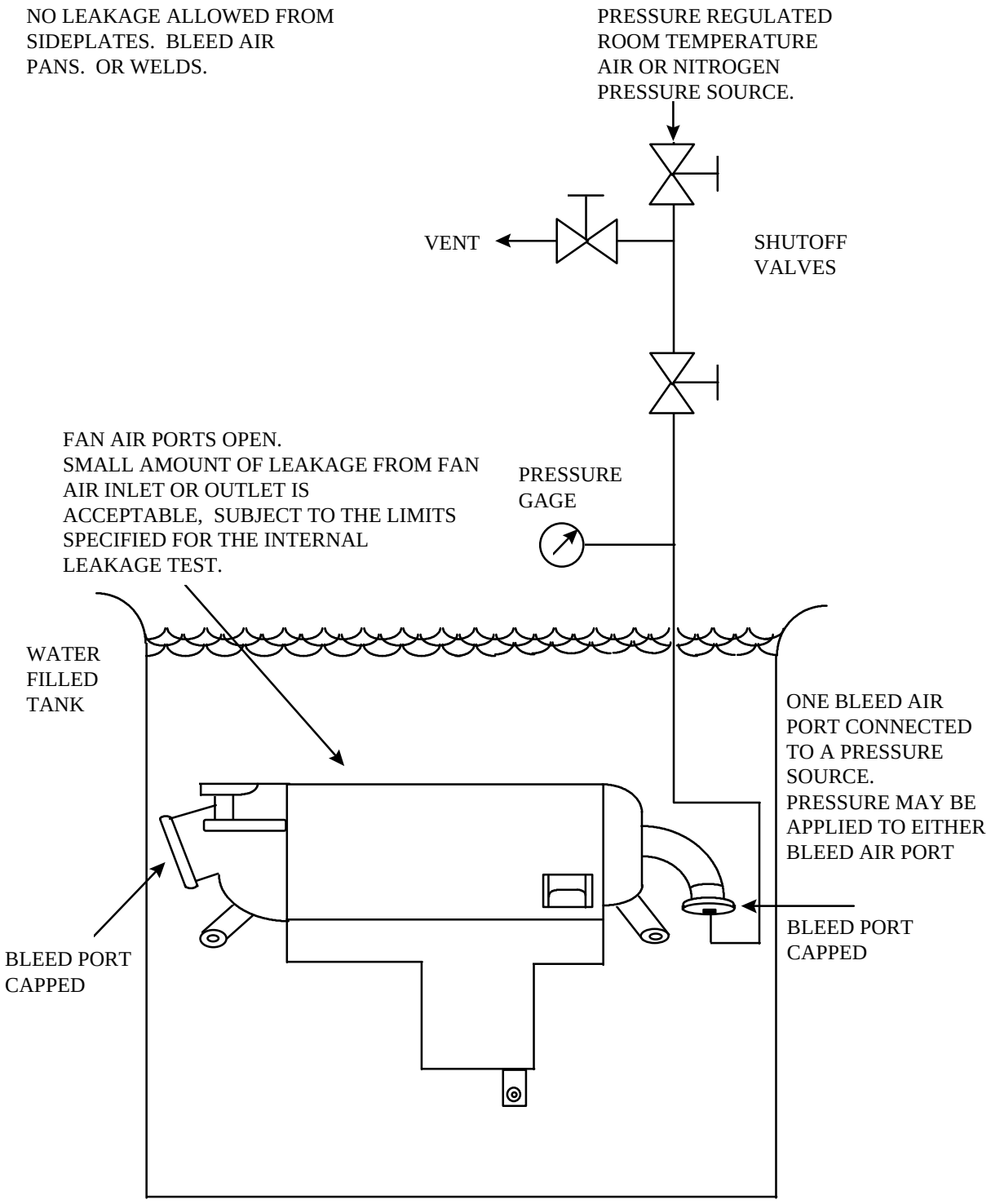


FIGURE 4. Leakage test setup.

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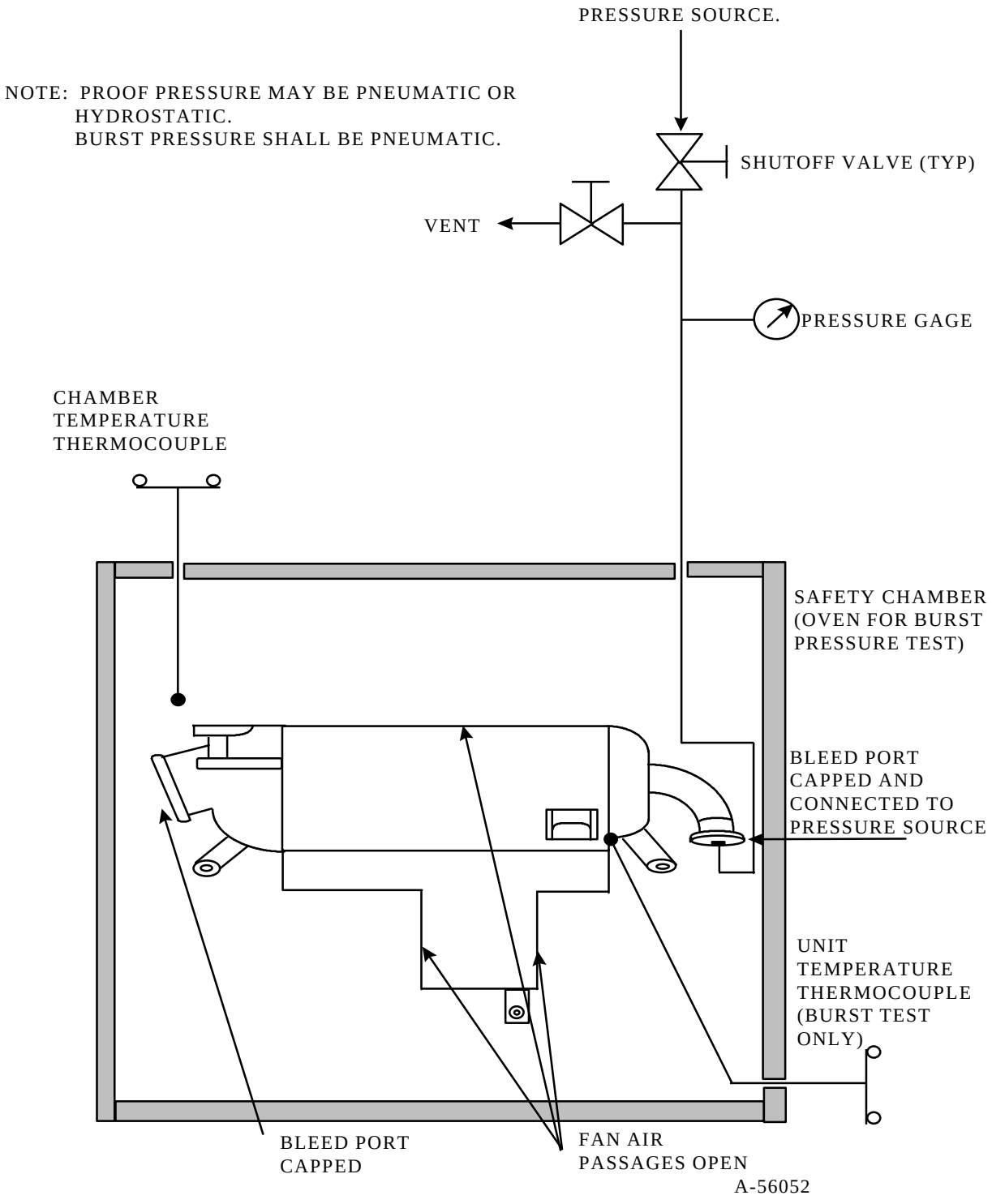


FIGURE 5. Proof and burst pressure test setup.

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